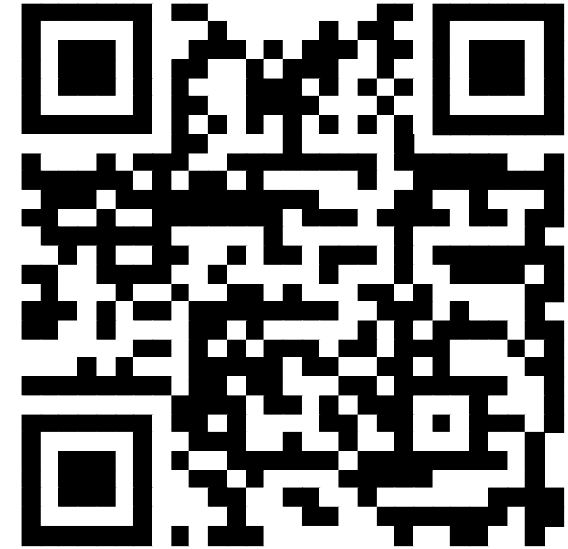


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In pandas DataFrames, data types are tied to

The rows

0%

The columns

0%

The whole dataframe

0%



##/##

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Preparing Results



In pandas DataFrames, data types are tied to

The rows



2%

The columns



97%

The whole dataframe



2%

RESULTS SLIDE



Let $a = \text{pd.Series}([1,2,3], \text{index}=[1,2,3])$ and $b = \text{pd.Series}([1,2,3], \text{index}=[1,3,2])$. What is $a + b$?

[1,2,3]

0%

[2,4,6]

0%

[2,5,5]

0%



Let `a = pd.Series([1,2,3], index=[1,2,3])` and `b = pd.Series([1,2,3], index=[1,3,2])`. What is `a + b`?

[1,2,3]

0%

[2,4,6]

3%

[2,5,5]



97%

RESULTS SLIDE



Let $a = \text{pd.Series}([1,2,3])$ and $b = \text{pd.Series}([1,2,3], \text{index}=[1,2,3])$. What is $a + b$?

[2,4,6]

0%

[1,2,3,1,2,3]

0%

[NaN,3,5,NaN]

0%



##/##

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Preparing Results



Let `a = pd.Series([1,2,3])` and `b = pd.Series([1,2,3], index=[1,2,3])`. What is `a + b`?

`[2,4,6]`

49%

`[1,2,3,1,2,3]`

0%

`[NaN,3,5,NaN]`

51%

RESULTS SLIDE



##/##

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Question slide



Let `a = pd.Series([1,2,3])`. What is `a + a[::-1]`?

[2,4,6]

0%

[4,4,4]

0%

[1,2,3,3,2,1]

0%



74

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Preparing Results



Let `a = pd.Series([1,2,3])`. What is `a + a[::-1]`?

[2,4,6]



3%

[4,4,4]



95%

[1,2,3,3,2,1]



3%

RESULTS SLIDE

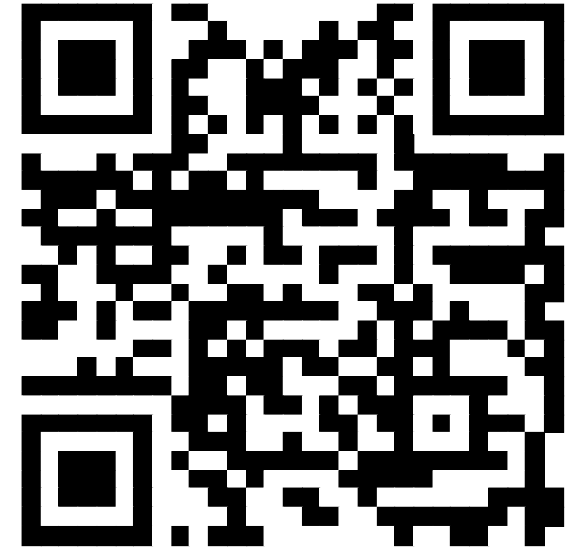
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What column is the best candidate for recoding as a categorical variable?

a

0%

b

0%

c

0%

```
>>> summarize_columns(df)
   name  dtype  unique  size (MB)
0    a  float64    5238        48
1    b  float64     16        48
2    c  float64  2137286       48
Total size: 144 MB
```



62

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Preparing Results

What column is the best candidate for recoding as a categorical variable?

a

0%

b

100%

c

0%

```
>>> summarize_columns(df)
   name  dtype  unique  size (MB)
0    a float64    5238         48
1    b float64      16         48
2    c float64 2137286         48
Total size: 144 MB
```

RESULTS SLIDE



Can we safely convert column b to int8?

Yes

0%

No

0%

Can't say

0%

```
>>> summarize_columns(df)
   name  dtype  unique  size (MB)
0    a  int64    5238         48
1    b  int64      16         48
2    c  int64 2137286         48
Total size: 144 MB
```



72/10

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Preparing Results



Can we safely convert column b to int8?

Yes



22%

No



3%

Can't say



75%

```
>>> summarize_columns(df)
   name  dtype  unique  size (MB)
0    a  int64    5238      48
1    b  int64      16      48
2    c  int64  2137286      48
Total size: 144 MB
```

RESULTS SLIDE

**Assume we work with float16. Let a = 100 and b = 0.1.
What is a + b?**

100

0%

100.1

0%

0.1

0%

Floating point (fractional numbers)

	resolution	min/max
float16	0.001	$\pm 6.55 * 10^4$
float32	1e-6	$\pm 3.40 * 10^{38}$
float64	1e-15	$\pm 1.80 * 10^{308}$



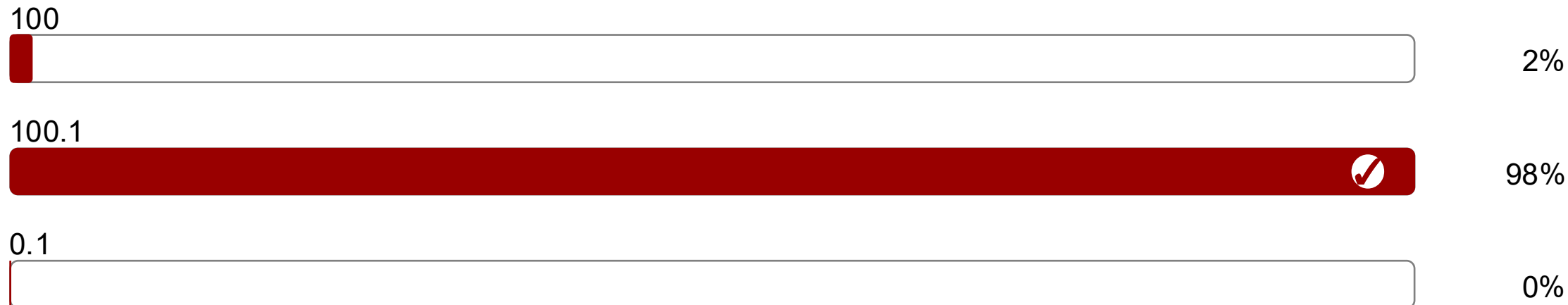
##/##

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Preparing Results

Assume we work with float16. Let $a = 100$ and $b = 0.1$.
What is $a + b$?



Floating point (fractional numbers)

	resolution	min/max
float16	0.001	$\pm 6.55 \cdot 10^4$
float32	1e-6	$\pm 3.40 \cdot 10^{38}$
float64	1e-15	$\pm 1.80 \cdot 10^{308}$

RESULTS SLIDE

**Assume we work with float16. Let $a = 1000$ and $b = 0.1$.
What is $a + b$?**

1000

0%

1000.1

0%

0.1

0%

Floating point (fractional numbers)

	resolution	min/max
float16	0.001	$\pm 6.55 * 10^4$
float32	1e-6	$\pm 3.40 * 10^{38}$
float64	1e-15	$\pm 1.80 * 10^{308}$



Assume we work with float16. Let $a = 1000$ and $b = 0.1$.
What is $a + b$?

1000



84%

1000.1



16%

0.1



0%

Floating point (fractional numbers)

	resolution	min/max
float16	0.001	$\pm 6.55 \cdot 10^4$
float32	1e-6	$\pm 3.40 \cdot 10^{38}$
float64	1e-15	$\pm 1.80 \cdot 10^{308}$

RESULTS SLIDE



Assume we work with float16. Let $a = 1000$. What is $a * a$?

1000

0%

1000000

0%

Inf

0%

Floating point (fractional numbers)

	resolution	min/max
float16	0.001	$\pm 6.55 * 10^4$
float32	1e-6	$\pm 3.40 * 10^{38}$
float64	1e-15	$\pm 1.80 * 10^{308}$



65

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Preparing Results



Assume we work with float16. Let $a = 1000$. What is $a * a$?

1000



12%

1000000



48%

Inf



40%

Floating point (fractional numbers)

	resolution	min/max
float16	0.001	$\pm 6.55 * 10^4$
float32	1e-6	$\pm 3.40 * 10^{38}$
float64	1e-15	$\pm 1.80 * 10^{308}$

RESULTS SLIDE



Assume we work with float32. Let $a = 0.1$. What is $a^{**} 8$?

0.0

0%

1.0e-8

0%

NaN

0%

Floating point (fractional numbers)

	resolution	min/max
float16	0.001	$\pm 6.55 * 10^4$
float32	1e-6	$\pm 3.40 * 10^{38}$
float64	1e-15	$\pm 1.80 * 10^{308}$



##/##

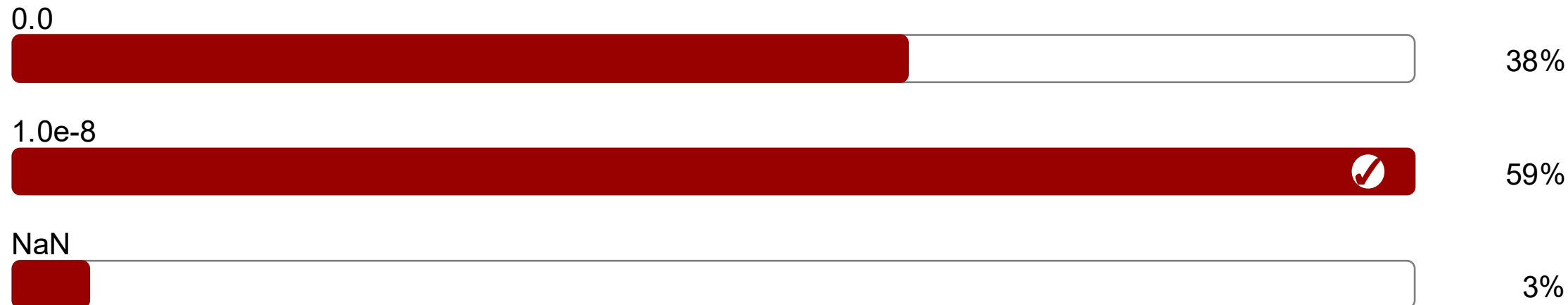
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Preparing Results



Assume we work with float32. Let $a = 0.1$. What is $a^{**} 8$?



Floating point (fractional numbers)

	resolution	min/max
float16	0.001	$\pm 6.55 \cdot 10^4$
float32	1e-6	$\pm 3.40 \cdot 10^{38}$
float64	1e-15	$\pm 1.80 \cdot 10^{308}$

RESULTS SLIDE

**Assume we work with float32. Let $a = 0.1$ and $b = 1.0$.
What is $b + a^{**8}$?**

- 1.0

0%
- 1.00000001

0%
- NaN

0%

Floating point (fractional numbers)

	resolution	min/max
float16	0.001	$\pm 6.55 * 10^4$
float32	1e-6	$\pm 3.40 * 10^{38}$
float64	1e-15	$\pm 1.80 * 10^{308}$



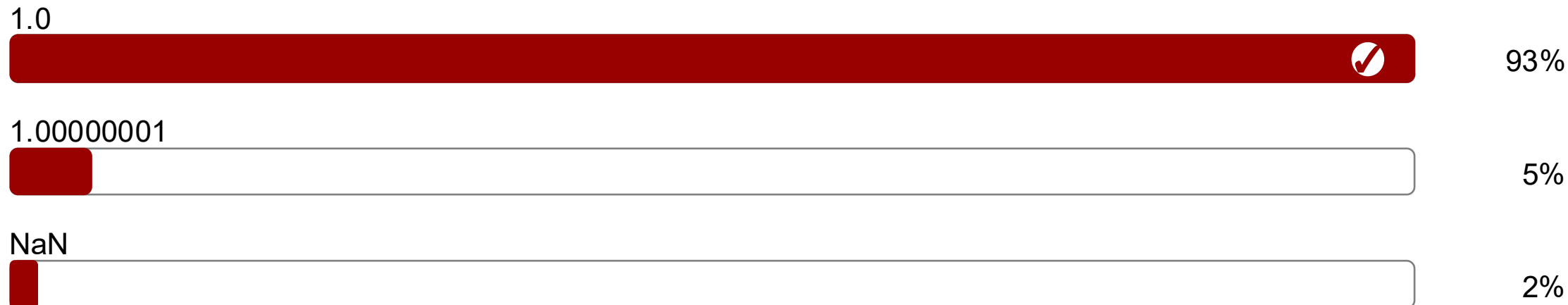
##/##

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Preparing Results

Assume we work with float32. Let $a = 0.1$ and $b = 1.0$.
What is $b + a^{**} 8$?



Floating point (fractional numbers)

	resolution	min/max
float16	0.001	$\pm 6.55 \cdot 10^4$
float32	1e-6	$\pm 3.40 \cdot 10^{38}$
float64	1e-15	$\pm 1.80 \cdot 10^{308}$

RESULTS SLIDE

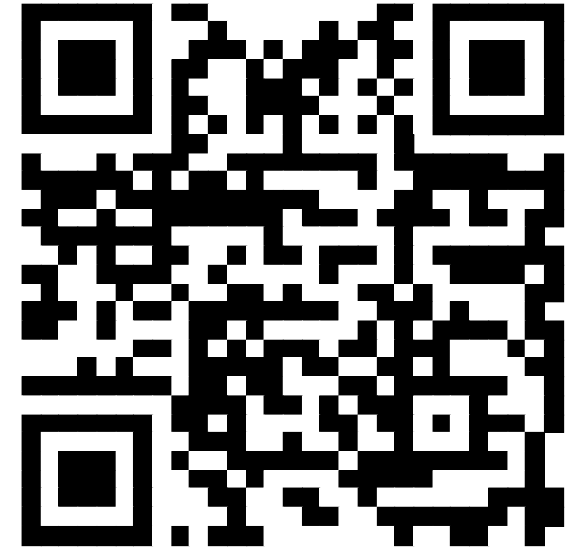
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How does PyArrow handle missing data?

Convert to float and use NaNs

0%

Automatically removes rows with missing data

0%

Extra data structure

0%



41

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Preparing Results



How does PyArrow handle missing data?

Convert to float and use NaNs



7%

Automatically removes rows with missing data



2%

Extra data structure



90%

RESULTS SLIDE



Given the number 0.123456789, how many bytes does it take to store as text? (Assume 1 byte per char)

8

0%

10

0%

11

0%



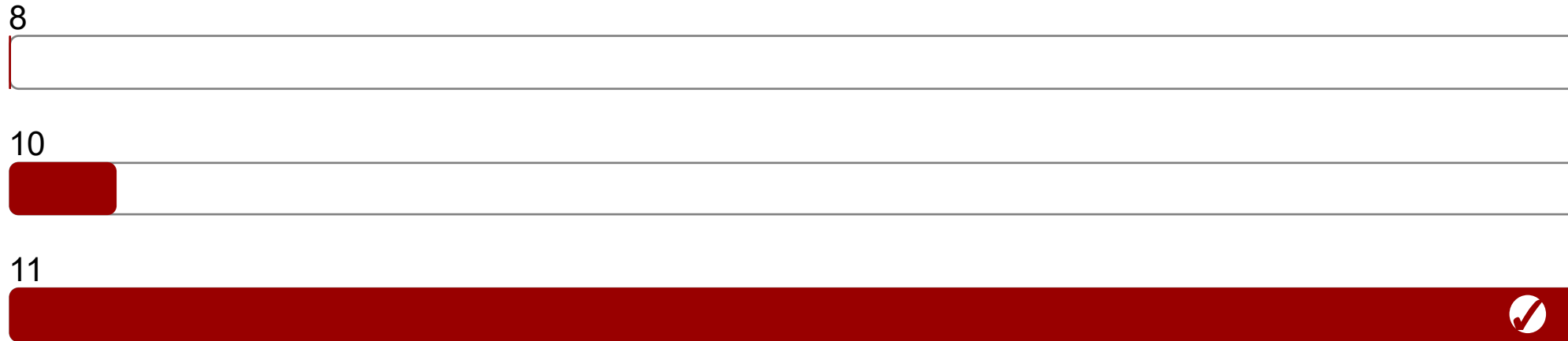
##/##

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Preparing Results

Given the number 0.123456789, how many bytes does it take to store as text? (Assume 1 byte per char)



RESULTS SLIDE



Given the time 2024-03-12 08:30, how many bytes does it take to store as text? (Assume 1 byte per char)

8

0%

16

0%

32

0%



Given the time 2024-03-12 08:30, how many bytes does it take to store as text? (Assume 1 byte per char)

8

0%

16

100%

32

0%

RESULTS SLIDE

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